Srimathi Ravisankar

Email : srimathi.ravisankar18@gmail.com
linkedin.com/in/srimathi-ravisankar | GitHub | LeetCode
Mobile : +1 608-395-1960

Education

University of Wisconsin–Madison — Madison, USA
Master of Science in Computer Science; GPA: 3.9/4.0 — *Aug 2025 – May 2027*

Madras Institute of Technology — Chennai, India
Bachelor of Engineering in Computer Science; CGPA: 9.4/10 — *Aug 2018 – May 2022*

Coursework: *Big Data Systems, High-Performance Computing, Machine Learning, AI Agents, Database Systems, Operating Systems, Parallel & Distributed Systems, Computer Architecture, Deep learning, Data structures & Object-Oriented Programming*

Experience

University of Wisconsin-Madison — USA
Graduate Teaching Assistant | Machine Organization and C Programming — *Aug 2025 – May 2027*

Microsoft — USA
SWE / AI Intern | Python, C++, C#, Android XR, Azure, CI/CD, RAG, Pinecone — *May – Jul 2026*

- Enabled Microsoft **Mixed-Reality Windows App** streaming, built for **Meta Quest 3** headsets, to run end-to-end on **Android XR Emulator**, enabling development and testing of mixed-reality apps without physical XR hardware.
- Engineered the emulator's video streaming pipeline using **CPU frame buffers**, multithreaded frame processing, and **buffer reuse** to minimize memory-copy overhead, and established **TCP** streaming between the Windows host and virtual XR device.
- Built E2E tests and integrated them into **Azure DevOps CI/CD**, enabling scalable validation without dedicated headset & lab infrastructure, improving developer productivity by 30% and reduced the cost of physical test-lab hardware by 75%.
- Developed an **AI Agent** using **RAG**, **vector embeddings** & **dependency graphs** for **regression test selection**, running only tests impacted by code changes and reducing CI execution time from 30 to 10 mins; implemented an **LLM auto-fix** loop to diagnose and repair failing test cases.

Barclays — India
*Software Developer (**3 YOE**) | Java, Spring Boot, Oracle, Docker, Kubernetes, Angular, GraphQL* — *Aug 2022 – Aug 2025*

- Developed **REST APIs** and **microservices** backend for Barclays' Corporate digital banking platform, aggregating transaction and account data across 8+ core payment systems for 30K+ corporate users to serve over 50K+ daily transactions.
- Accelerated data archival pipeline using **multithreaded batch processing**, **indexing** & **partitioning** ∼100K transactions into parallel 10K batches, reducing runtime from 24 hours to 5 minutes.
- Optimized a centralized data service by implementing **in-memory caching** with **GZip** compression, cutting payload size by 65%, database I/O by 45%, and overall latency from 200 ms to 65 ms with **70% cache hit rate**.
- Optimized **Spring Data JPA** query execution times on **Oracle** from **200–500 ms** to under **10 ms** by designing **composite-key indexes** across **1M+** record tables.
- Built an incident triage **multi-agent system** using **RAG**, code-graph traversal, and memory-augmented reasoning to localize root causes and generate patches, correctly localizing 82 of 100 historical incidents.

Samsung — India
SDE Intern | Node.js, React, Express, RabbitMQ, Elasticsearch, CockroachDB, PostgreSQL, Kibana, Redis — *Mar – Jul 2022*

- Developed a real-time full stack **Payment Monitoring Dashboard** for Samsung US Ecommerce to process and visualize **15,000+ daily transactions** across payment partners like Visa, PayPal, and Samsung Pay.

Citibank — India
SDE Intern | Java, Lucene, Amazon S3, Kafka — *Mar – Jul 2021*

- Built a fault-tolerant distributed search service for **2,000+ S3** compliance documents with sharding, replication and automated failover, using multithreaded ingestion to raise indexing throughput by **65%** at 1,500+ QPS.

Projects

Distributed Build & Regression System | *Go, gRPC, C++, CMake, Python, React, Prometheus, Grafana* — *Jun – Oct 2026*

- Built an **LSF-style** build/regression scheduler in **Go** that dispatches **C++/CMake** builds and **GoogleTest** suites over **gRPC** to distributed workers with **test sharding**, **build caching** and **failover**, plus a **React** dashboard and **Prometheus** metrics to track and triage runs.

Fault-Tolerant Distributed Key-Value Store | *Go, gRPC, Protobuf, Raft* — *Aug – Dec 2025*

- Built a strongly consistent, linearizable KV store in **Go** using the **Raft consensus protocol**, implementing leader election, **gRPC-based log replication**, snapshot compaction, and automatic recovery from node failures and network partitions.

Distributed LLM Inference Platform | *Go, Python, C++, Kubernetes, CUDA* — *Aug – Dec 2025*

- Engineered a distributed LLM serving engine with **continuous batching**, **PagedAttention KV-cache management**, and custom **GPU-aware Kubernetes scheduling**, boosting throughput by **25%** and reducing P99 tail latency.

Technical Skills

- **Languages**: C++, Golang, Python, C, Java, Perl, Tcl, Bash/Shell, JavaScript, TypeScript, C#, SQL, Rust
- **Frameworks**: Spring Boot, TensorFlow, PyTorch, Keras, Node.js, Express, Flask, Angular, React, LangGraph, CUDA
- **Big Data & Databases**: MySQL, PostgreSQL, Oracle, MongoDB, Spark, Redis, Kafka, RabbitMQ, CockroachDB
- **Tools & Cloud**: Linux, Jenkins CI/CD, Azure DevOps, Docker, Kubernetes, AWS, Prometheus, Grafana, Android Studio, JIRA
- **Build & Quality Tools**: Git, Perforce, CMake, Bazel, GoogleTest, gdb, Valgrind, AddressSanitizer, clang-tidy, gcov